

Du 17 au 26 Avril 2026

CAHIER DES CHARGES

Mise en place d'un système de
détection et prévention
d'intrusions – SNORT (IDS/IPS)

Laureen RUELLE
MFR DE ST EGREVE

SOMMAIRE

1. Cahier des charges – Expression des besoins.....	3
1.1. Descriptif de l'existant	3
1.2. Besoin	3
1.3. Contraintes	3
1.3.1. Temps.....	3
1.3.2. Budget	3
1.3.3. Juridique	3
1.3.4. Organisation	3
2. Analyse	4
2.1. Descriptifs des solutions	4
2.2. Comparaison des solutions.....	4
2.3. Choix d'une solution – Argumentation	4
2.4. Etude de l'impact sur le SI existant	4
2.4.1. Sécurité	4
2.4.2. Performance	4
2.4.3. Juridique	4
2.4.4. Ergonomie	4
2.5. Prévision des tests de validation, à quel moment seront effectués les tests.	4
2.6. Déploiement.....	5
2.6.1. Procédure	5
2.6.2. Pilotes	5
2.6.3. Horaires.....	5
3. Mise en place	6
3.1. Méthodologie (pas un tuto !)	6
3.2. Les procédures spécifiques seront documentées dans des liens externes à ce document.....	6
3.3. Configuration.....	6
3.4. Réalisation des tests.....	6
3.5. Rapport de test et de déploiement.....	6
4. Bilan	7
Annexe – Procédure : Mise en place de SNORT sur Pfsense.....	8
1. Mode IDS uniquement (détection uniquement)	8
2. Passage en IPS	19

3.	Schéma de ce qu'il a été mis en place	23
-----------	---	----

1. Cahier des charges – Expression des besoins

1.1. Descriptif de l'existant

Le système d'information actuel repose sur :

- Un pare-feu pfSense assurant le filtrage réseau
- Absence de système de détection/prévention d'intrusion (IDS/IPS)
- Supervision limitée aux logs du firewall

Limite principale : absence de visibilité sur les attaques réseau avancées.

1.2. Besoin

Mettre en place une solution permettant :

- La détection des intrusions (IDS)
- La prévention des attaques (IPS)
- L'analyse du trafic en temps réel
- L'alerte et la journalisation des événements de sécurité

1.3. Contraintes

1.3.1. Temps

Déploiement souhaité : 1 à 2 semaines

Tests inclus dans le planning

1.3.2. Budget

- Solution open source privilégiée (coût logiciel nul)
- Coût possible :
 - Abonnement règles Snort (optionnel)
 - Ressources matérielles supplémentaires

1.3.3. Juridique

- Respect du RGPD (analyse du trafic réseau)
- Journalisation conforme (durée de conservation des logs)

1.3.4. Organisation

- Intervention de l'équipe IT / sécurité

2. Analyse

2.1. Descriptifs des solutions

2.2. Comparaison des solutions

2.3. Choix d'une solution – Argumentation

Snort est retenu car :

- Intégration native dans pfSense
- Large base de règles
- Bonne documentation

2.4. Etude de l'impact sur le SI existant

2.4.1. Sécurité

- Amélioration significative de la détection d'attaques
- Réduction des risques d'intrusion

2.4.2. Performance

- Augmentation de la charge CPU
- Nécessité d'ajuster les règles

2.4.3. Juridique

- Surveillance du trafic = potentiellement sensible
- Obligation de transparence

2.4.4. Ergonomie

- Interface Pfsense intuitive

2.5. Prévision des tests de validation, à quel moment seront effectués les tests.

- Après configuration initiale
- Après chaque modification majeure

2.6. Déploiement

2.6.1. Procédure

Mode IDS uniquement :

- Installation du package Snort
- Configuration des interfaces
- Chargement des règles
- Activation IDS
- Tests

Puis mode IPS :

- Observer : mode IDS, analyse des alertes
- Nettoyer les règles
- Créer une « Whitelist »
- Choisir le bon mode
- Test

2.6.2. Pilotes

- Administrateur réseau

2.6.3. Horaires

- Déploiement hors production (soir/week-end)
- Tests en heures creuses

3. Mise en place

3.1. Méthodologie (pas un tuto !)

- Approche progressive :
 - Mode IDS (monitoring)
 - Analyse des faux positifs
 - Passage en mode IPS

3.2. Les procédures spécifiques seront documentées dans des liens externes à ce document

Voir Annexe à la fin de la documentation

3.3. Configuration

Configuration principale :

- Interfaces surveillées (WAN/LAN)
- Activation des règles communautaires
- Configuration des alertes
- Mise en place de whitelist/blacklist

3.4. Réalisation des tests

Tests effectués :

- Scan réseau (ex : nmap)
- Simulation d'attaque
- Vérification des logs

Livrables :

- Rapport de tests
- Analyse des faux positifs
- Recommandations

3.5. Rapport de test et de déploiement

Voir Annexe à la fin de la documentation

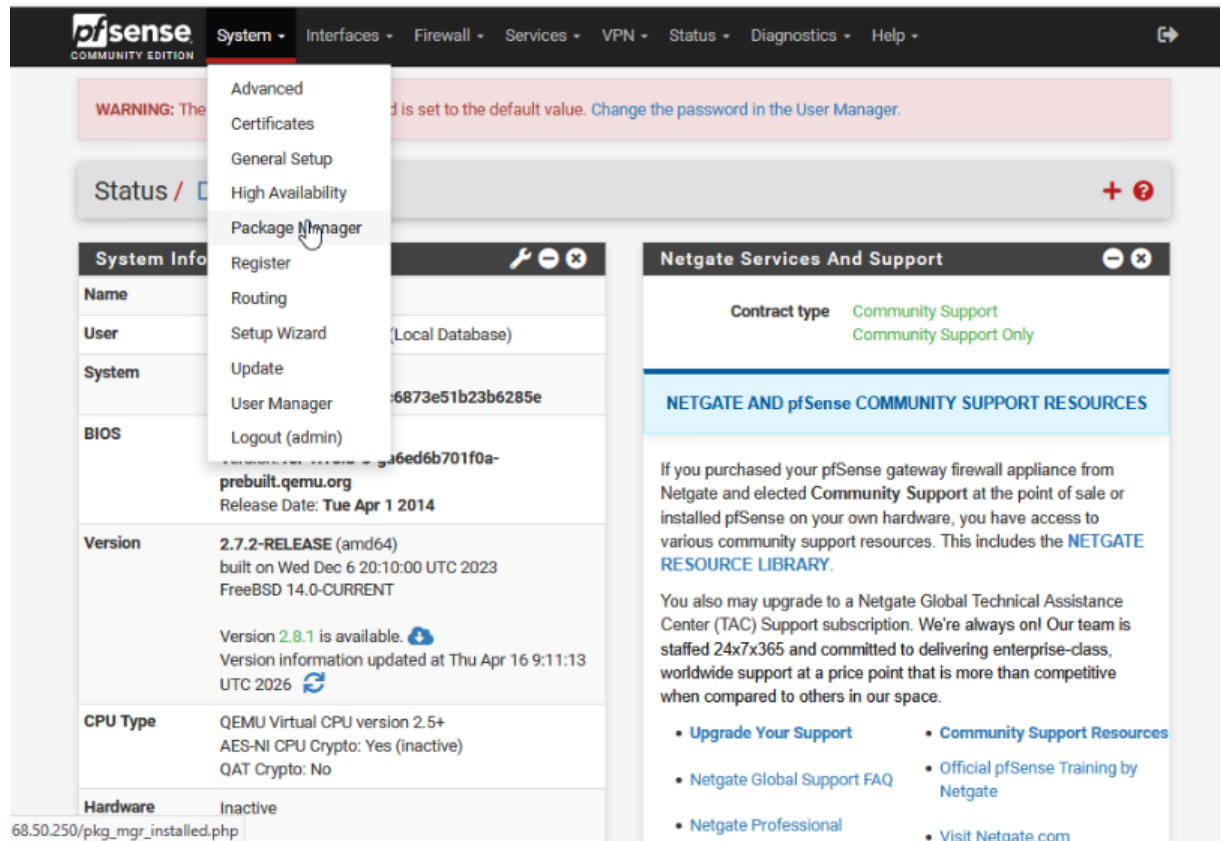
4. Bilan

La mise en place d'IDS/IPS est réussie. Cela permet une amélioration de la sécurité globale et une surveillance du réseau renforcée.

Annexe – Procédure : Mise en place de SNORT sur Pfsense

1. Mode IDS uniquement (détection uniquement)

Etape 1 : Installer le package Snort



The screenshot shows the pfSense web interface. The 'System' menu is open, and 'Package Manager' is selected. The 'System Info' panel on the left displays details about the system, including the version (2.7.2-RELEASE) and hardware. The 'Netgate Services And Support' panel on the right provides information about the support contract and resources.

System Info

Field	Value
Name	prebuilt.qemu.org
User	6873e51b23b6285e
System	6873e51b23b6285e
BIOS	6873e51b23b6285e
Version	2.7.2-RELEASE (amd64) built on Wed Dec 6 20:10:00 UTC 2023 FreeBSD 14.0-CURRENT Version 2.8.1 is available. Version information updated at Thu Apr 16 9:11:13 UTC 2026
CPU Type	QEMU Virtual CPU version 2.5+ AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware	Inactive

Netgate Services And Support

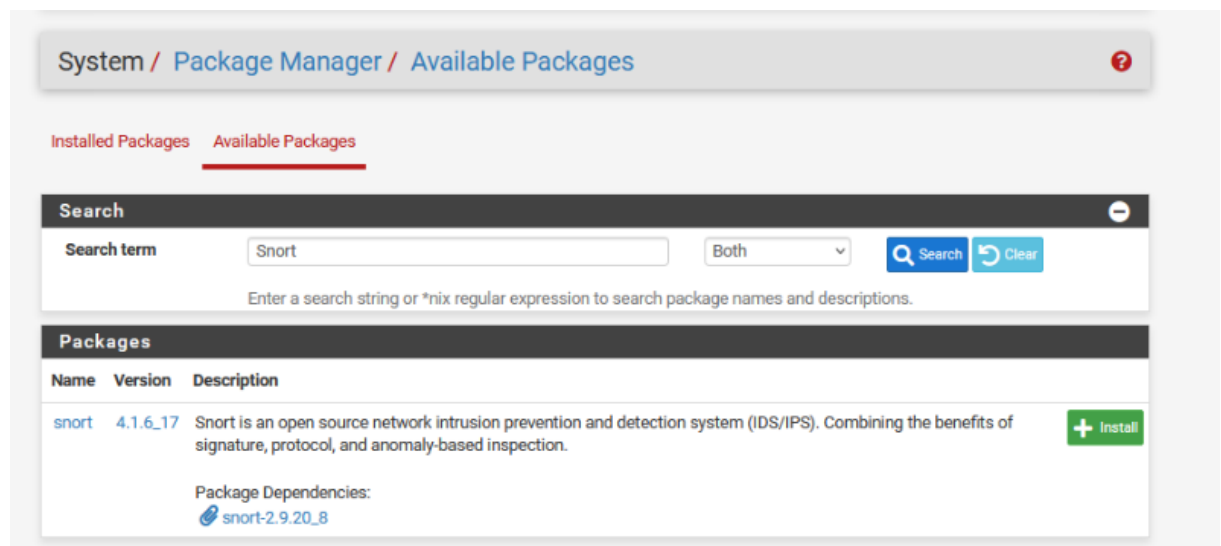
Contract type: Community Support
Community Support Only

NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the [NETGATE RESOURCE LIBRARY](#).

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- Upgrade Your Support
- Community Support Resources
- Netgate Global Support FAQ
- Official pfSense Training by Netgate
- Netgate Professional
- Visit Netgate.com



The screenshot shows the 'System / Package Manager / Available Packages' page. The 'Available Packages' tab is selected. A search bar is present, and the search term 'Snort' is entered. The search results show a table of available packages, with 'snort' listed as the first result. The package description and dependencies are also displayed.

System / Package Manager / Available Packages

Installed Packages Available Packages

Search

Search term: Snort Both Search Clear

Enter a search string or *nix regular expression to search package names and descriptions.

Packages

Name	Version	Description	Action
snort	4.1.6_17	Snort is an open source network intrusion prevention and detection system (IDS/IPS). Combining the benefits of signature, protocol, and anomaly-based inspection.	+ Install

Package Dependencies:
snort-2.9.20_8

System / Package Manager / Installed Packages

Installed Packages
Available Packages

Installed Packages

Name	Category	Version	Description	Actions
✓ openvpn-client-export	security	1.9.2	Exports pre-configured OpenVPN Client configurations directly from pfSense software. Package Dependencies: openvpn-client-export-2.6.7 openvpn-2.6.8_1 zip-3.0_1 7-zip-23.01	
✓ snort	security	4.1.6.17	Snort is an open source network intrusion prevention and detection system (IDS/IPS). Combining the benefits of signature, protocol, and anomaly-based inspection. Package Dependencies: snort-2.9.20.8	

= Update
 = Current

 = Remove
 = Information
 = Reinstall

Newer version available

Package is configured but not (fully) installed or deprecated

COMMUNITY EDITION

System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help

WARNING: The 'admin' account password is set to the default password in the User Manager.

System / Package Manager / Installed Packages

Installed Packages
Available Packages

Installed Packages

Name	Category	Version	Description	Actions
✓ openvpn-client-export	security	1.9.2	Exports pre-configured OpenVPN Client configurations directly from pfSense software. Package Dependencies: openvpn-client-export-2.6.7 openvpn-2.6.8_1 zip-3.0_1 7-zip-23.01	
✓ snort	security	4.1.6.17	Snort is an open source network intrusion prevention and detection system (IDS/IPS). Combining the benefits of signature, protocol, and anomaly-based inspection. Package Dependencies: snort-2.9.20.8	

= Update
 = Current

 = Remove
 = Information
 = Reinstall

Newer version available

Package is configured but not (fully) installed or deprecated

Auto Config Backup
Captive Portal
DHCP Relay
DHCP Server
DHCPv6 Relay
DHCPv6 Server
DNS Forwarder
DNS Resolver
Dynamic DNS
IGMP Proxy
NTP
PPPoE Server
Router Advertisement
SNMP
Snort
UPnP & NAT-PMP
Wake-on-LAN

9/23

Etape 2 : Configurer une interface

Services / Snort / Interfaces ?

Snort Interfaces Global Settings Updates Alerts Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync

Interface Settings Overview

Interface	Snort Status	Pattern Match	Blocking Mode	Description	Actions
-----------	--------------	---------------	---------------	-------------	---------

+ Add

Services / Snort / WAN - Interface Settings ?

Snort Interfaces Global Settings Updates Alerts Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync

WAN Settings

General Settings

Enable	☒ Enable interface
Interface	WAN (vtnet0) ▼ Choose the interface where this Snort instance will inspect traffic.
Description	WAN Enter a meaningful description here for your reference.
Snap Length	1518 ▼ Enter the desired interface snaplen value in bytes. Default is 1518 and is suitable for most applications.

Alert Settings

Send Alerts to System Log	☐ Snort will send Alerts to the firewall's system log. Default is Not Checked.
Enable Packet Captures	☐ Checking this option will automatically capture packets that generate a Snort alert into a tcpdump compatible file
Enable Unified2 Logging	☐ Checking this option will cause Snort to simultaneously log alerts to a unified2 binary format log file in the logging subdirectory for this interface. Default is Not Checked. Log size and retention limits for the Unified2 log should be configured on the LOG MGMT tab when this option is enabled.

Block Settings

Block Offenders	☐ Checking this option will automatically block hosts that generate a Snort alert. Default is Not Checked.
-----------------	---

Detection Performance Settings

Search Method	AC-BNFA ▼ Choose a fast pattern matcher algorithm. Default is AC-BNFA.
Split ANY-ANY	☐ Enable splitting of ANY-ANY port group. Default is Not Checked.
Search Optimize	☐ Enable search optimization. Default is Not Checked.
Stream Inserts	☐ Do not evaluate stream inserted packets against the detection engine. Default is Not Checked.
Checksum Check Disable	☐ Disable checksum checking within Snort to improve performance. Default is Not Checked.

Choose the Networks Snort Should Inspect and Whitelist

Home Net

default

View List

Choose the Home Net you want this interface to use.

Default Home Net adds only local networks, WAN IPs, Gateways, VPNs and VIPs.
Create an Alias to hold a list of friendly IPs that the firewall cannot see or to customize the default Home Net.

External Net

default

View List

Choose the External Net you want this interface to use.

External Net is networks that are not Home Net. Most users should leave this setting at default.
Create a Pass List and add an Alias to it, and then assign the Pass List here for custom External Net settings.

Choose a Suppression or Filtering List (Optional)

Alert Suppression and Filtering

default

View List

Choose the suppression or filtering file you want this interface to use.

Custom Configuration Options

Advanced Configuration Pass-Through

Enter any additional configuration parameters to add to the Snort configuration here, separated by a newline

Save

Services / Snort / Interfaces

Snort Interfaces

Global Settings

Updates

Alerts

Blocked

Pass Lists

Suppress

IP Lists

SID Mgmt

Log Mgmt

Sync

Interface Settings Overview

Interface	Snort Status	Pattern Match	Blocking Mode	Description	Actions
☐ WAN (vtnet0)	✓ ↻ 🔒	AC-BNFA	DISABLED	WAN	✎ 📄 🗑

+ Add

🗑 Delete

Snort Subscriber Rules

Enable Snort VRT ☐ Click to enable download of Snort free Registered User or paid Subscriber rules

[Sign Up for a free Registered User Rules Account](#)
[Sign Up for paid Snort Subscriber Rule Set \(by Talos\)](#)

Snort GPLv2 Community Rules

Enable Snort GPLv2 ☐ Click to enable download of Snort GPLv2 Community rules

The Snort Community Ruleset is a GPLv2 Talos certified ruleset that is distributed free of charge without any Snort Subscriber License restrictions. This ruleset is updated daily and is a subset of the subscriber ruleset.

Emerging Threats (ET) Rules

Enable ET Open ☒ Click to enable download of Emerging Threats Open rules

ETOpen is an open source set of Snort rules whose coverage is more limited than ETPro.

Enable ET Pro ☐ Click to enable download of Emerging Threats Pro rules

[Sign Up for an ETPro Account](#)
ETPro for Snort offers daily updates and extensive coverage of current malware threats.

Sourcefire OpenAppID Detectors

Enable OpenAppID ☐ Click to enable download of Sourcefire OpenAppID Detectors

The OpenAppID Detectors package contains the application signatures required by the AppID preprocessor and the OpenAppID text rules.

OpenAppID Version

Enable AppID Open Text Rules ☐ Click to enable download of the AppID Open Text Rules

Note - the AppID Open Text Rules file is maintained by a volunteer contributor and hosted by the pfSense team. The URL for the file is https://files.netgate.com/openappid/appid_rules.tar.gz.

FEODO Tracker Botnet C2 IP Rules

Enable FEODO Tracker Botnet C2 IP Rules ☐ Click to enable download of FEODO Tracker Botnet C2 IP rules

Feodo Tracker tracks certain families that are related to, or that evolved from, Feodo. Originally, Feodo was an ebanking Trojan used by cybercriminals to commit ebanking fraud. Since 2010, various malware families evolved from Feodo, such as Cridex, Dridex, Geodo, Heodo and Emotet.

Rules Update Settings

Update Interval

NEVER

Please select the interval for rule updates. Choosing NEVER disables auto-updates.

Update Start Time

00:28

Enter the rule update start time in 24-hour format (HH:MM). Default is 00 hours with a randomly chosen minutes value. Rules will update at the interval chosen above starting at the time specified here. For example, using a start time of 00:08 and choosing 12 Hours for the interval, the rules will update at 00:08 and 12:08 each day. The randomized minutes value should be retained to minimize the impact to the rules update site from large numbers of simultaneous requests.

Hide Depreciated Rules Categories

☐ Click to hide deprecated rules categories in the GUI and remove them from the configuration. Default is not checked.

Disable SSL Peer Verification

☐ Click to disable verification of SSL peers during rules updates. This is commonly needed only for self-signed certificates. Default is not checked.

General Settings

Remove Blocked Hosts Interval

NEVER

Please select the amount of time you would like hosts to be blocked. In most cases, one hour is a good choice.

Remove Blocked Hosts After Deinstall

☐ Click to clear all blocked hosts added by Snort when removing the package. Default is checked.

Keep Snort Settings After Deinstall

☒ Click to retain Snort settings after package removal.

Startup/Shutdown Logging

☐ Click to output detailed messages to the system log when Snort is starting and stopping. Default is not checked.

Save

3 nov

Etape 3 : Activer les règles

1. Installer les règles

Services → Snort → Updates

Services / Snort / Updates

Snort Interfaces

Global Settings

Updates

Alerts

Blocked

Pass Lists

Suppress

IP Lists

SID Mgmt

Log Mgmt

Sync

Installed Rule Set MD5 Signature

Rule Set Name/Publisher	MD5 Signature Hash	MD5 Signature Date
Snort Subscriber Ruleset	Not Enabled	Not Enabled
Snort GPLv2 Community Rules	Not Enabled	Not Enabled
Emerging Threats Open Rules	Not Downloaded	Not Downloaded
Snort OpenAppID Detectors	Not Enabled	Not Enabled
Snort AppID Open Text Rules	Not Enabled	Not Enabled
Feodo Tracker Botnet C2 IP Rules	Not Enabled	Not Enabled

Update Your Rule Set

Last Update

Unknown

Result: Unknown

Update Rules

Update Rules

Force Update

Activer Windows

Accédez aux paramètres pour Windows.

2. Cliquer sur « Update Rules »

Services → Snort → ton interface → Categories

Services / Snort / Interface Settings / WAN - Categories

Snort Interfaces Global Settings Updates Alerts Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync

WAN Settings WAN Categories WAN Rules WAN Variables WAN Preprocs WAN IP Rep WAN Logs

Automatic Flowbit Resolution

Resolve Flowbits ☒ If checked, Snort will auto-enable rules required for checked flowbits. Default is Checked.
Snort will examine the enabled rules in your chosen rule categories for checked flowbits. Any rules that set these dependent flowbits will be automatically enabled and added to the list of files in the interface rules directory.

Select the rulesets (Categories) Snort will load at startup

⬆ - Category is auto-enabled by SID Mgmt conf files
⬇ - Category is auto-disabled by SID Mgmt conf files

Select All Unselect All Save

ET Open rules have not been downloaded. Snort Subscriber rules are not enabled. Snort OPENAPPID rules are not enabled.

Save

3. Activer les règles sur l'interface

Services → Snort → WAN Categories

Snort Interfaces Global Settings Updates Alerts Blocked Pass Lists Suppress IP Lists SID Mgmt Log Mgmt Sync

WAN Settings WAN Categories WAN Rules WAN Variables WAN Preprocs WAN IP Rep WAN Logs

Automatic Flowbit Resolution

Resolve Flowbits ☒ If checked, Snort will auto-enable rules required for checked flowbits. Default is Checked.
Snort will examine the enabled rules in your chosen rule categories for checked flowbits. Any rules that set these dependent flowbits will be automatically enabled and added to the list of files in the interface rules directory.

Select the rulesets (Categories) Snort will load at startup

⬆ - Category is auto-enabled by SID Mgmt conf files
⬇ - Category is auto-disabled by SID Mgmt conf files

Select All Unselect All Save

Enable Ruleset: ET Open Rules Snort Subscriber rules are not enabled. Snort OPENAPPID rules are not enabled.

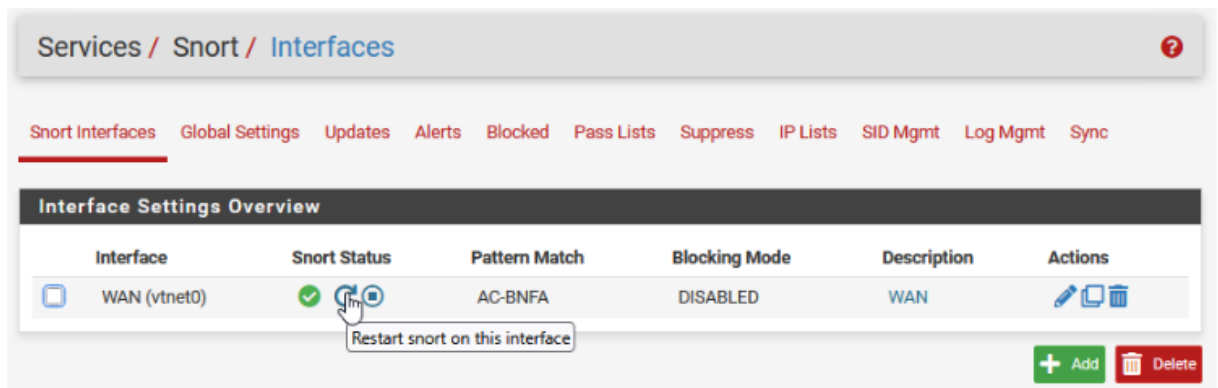
☐	emerging-activex.rules
☐	emerging-attack_response.rules
☐	emerging-botcc.portgrouped.rules
☐	emerging-botcc.rules
☐	emerging-chat.rules
☐	emerging-ciarmy.rules
☐	emerging-compromised.rules
☐	emerging-current_events.rules
☐	emerging-deleted.rules

Activer au minimum :

• Emerging-scan • Emerging-trojan • Emerging-malware • Emerging-web_server	• Analyse émergente • Cheval de Troie émergent • Logiciel malveillant émergent • Serveur web émergent
--	---

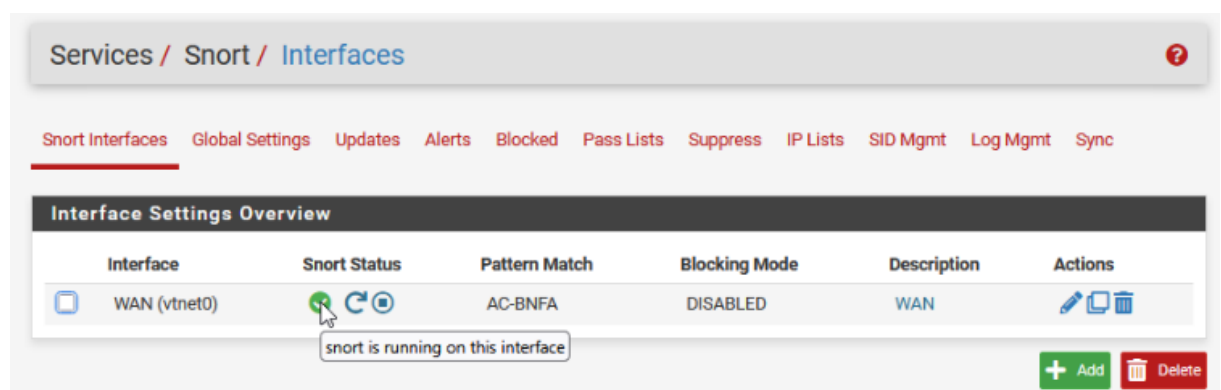
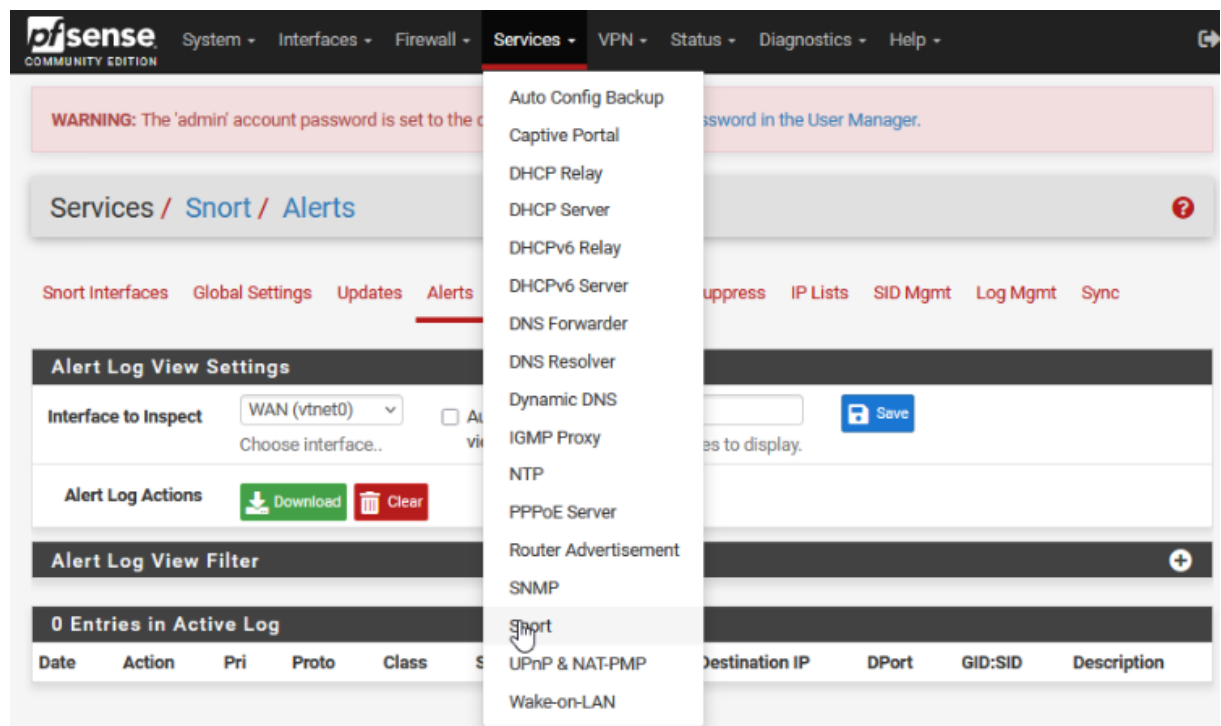
☒	emerging-malware.rules
☐	emerging-misc.rules
☐	emerging-mobile_malware.rules
☐	emerging-netbios.rules
☐	emerging-p2p.rules
☐	emerging-policy.rules
☐	emerging-pop3.rules
☐	emerging-retired.rules
☐	emerging-rpc.rules
☐	emerging-scada.rules
☒	emerging-scan.rules
☒	emerging-trojan.rules
☐	emerging-user_agents.rules
☐	emerging-voip.rules
☐	emerging-web_client.rules
☒	emerging-web_server.rules

4. Redémarrer SNORT



Etape 4 : Test si SNORT fonctionne

1. Vérification de l'état :



2. Vérification si le scan fonctionne.

Depuis une machine Cliente en interne :

```
PS C:\Users\svc_admin> nmap -sS -A scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-30 13:46 +0200
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
80/tcp    open  http           Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-favicon: Nmap Project
|_http-title: Go ahead and ScanMe!
443/tcp    closed https
3389/tcp   closed ms-wbt-server
Device type: general purpose|firewall|router
Running (JUST GUESSING): Linux 4.X|2.6.X|3.X|5.X (97%), IPFire 2.X (91%), MikroTik RouterOS 7.X (88%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3.10 cpe:/o:linux:linux_k
ernel:5.4 cpe:/o:ipfire:ipfire:2.27 cpe:/o:linux:linux_kernel:6.1 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5
.6.3
Aggressive OS guesses: Linux 4.19 - 5.15 (97%), Linux 2.6.32 or 3.10 (92%), Linux 4.0 - 4.4 (92%), Linux 5.4 (92%), Linu
x 2.6.32 (91%), Linux 4.15 (91%), IPFire 2.27 (Linux 5.15 - 6.1) (91%), Linux 2.6.32 - 2.6.35 (90%), Linux 2.6.32 - 2.6.
39 (90%), Linux 2.6.32 - 3.0 (88%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 22 hops

TRACEROUTE (using port 443/tcp)
HOP RTT      ADDRESS
1 2.00 ms   routeur (192.168.30.200)
2 2.00 ms   pfsense1.laureen.local (192.168.50.250)
3 2.00 ms   192.168.1.254
4 ...
5 10.00 ms  rennes-9k-1-lo10.intf.routers.proxad.net (194.149.162.250)
6 11.00 ms  prs-b17-link.ip.twelve99.net (213.248.91.224)
7 11.00 ms  prs-bb2-link.ip.twelve99.net (62.115.136.224)
8 ...
9 96.00 ms  rest-b2-link.ip.twelve99.net (62.115.121.216)
10 91.00 ms  ash-bb2-link.ip.twelve99.net (62.115.136.201)
11 96.00 ms  ae3.r21.iad02.mag.netarch.akamai.com (23.209.165.106)
12 97.00 ms  ae0.r21.iad02.icn.netarch.akamai.com (23.209.165.91)
13 ...
14 155.00 ms ae16.r02.sjc01.icn.netarch.akamai.com (23.193.113.29)
15 153.00 ms  ae1.r11.sjc01.iem.netarch.akamai.com (23.207.232.35)
16 151.00 ms  ae16.r02.sjc01.icn.netarch.akamai.com (23.193.113.29)
17 ... 21
22 155.00 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.59 seconds
```

Le scan Nmap a bien fonctionné mais cela ne prouve pas encore que Snort fonctionne.

Est-ce que Snort a détecté ce scan ?

Services → Snort → Alerts

80 Entries in Active Log										
Date	Action	Pri	Proto	Class	Source IP	SPort	Destination IP	DPort	GID:SID	Description
2026-04-30 11:47:08		3	TCP	Unknown Traffic	45.33.32.156  	80	192.168.1.250  	50405	120:3   	(http_inspect) NO CONTENT-LENGTH OR TRANSFER-ENCODING IN HTTP RESPONSE
2026-04-30 11:47:08		3	TCP	Unknown Traffic	192.168.1.250  	50405	45.33.32.156  	80	120:8   	(http_inspect) INVALID CONTENT-LENGTH OR CHUNK SIZE
2026-04-30 11:47:06		1	TCP	Web Application Attack	192.168.1.250  	15811	45.33.32.156  	80	1:2024364   	ET SCAN Possible Nmap User-Agent Observed
2026-04-30 11:47:06		1	TCP	Web Application Attack	192.168.1.250  	14854	45.33.32.156  	80	1:2024364   	ET SCAN Possible Nmap User-Agent Observed
2026-04-30 11:47:05		1	TCP	Web Application Attack	192.168.1.250  	65456	45.33.32.156  	80	1:2024364   	ET SCAN Possible Nmap User-Agent Observed
2026-04-30 11:47:05		1	TCP	Web Application Attack	192.168.1.250  	64211	45.33.32.156  	80	1:2024364   	ET SCAN Possible Nmap User-Agent Observed
2026-04-30 11:47:05		1	TCP	Web Application Attack	192.168.1.250  	2744	45.33.32.156  	80	1:2024364   	ET SCAN Possible Nmap User-Agent Observed
2026-04-30 11:47:04		1	TCP	Web Application	192.168.1.250  	25374	45.33.32.156  	80	1:2024364   	ET SCAN Possible Nmap User-Agent

Ceci justifie que le SCAN fonctionne.

2. Passage en IPS

Attention, le passage en IPS nécessite des réglages et des vérifications avant et le blocage demande une configuration propre.

Ce qu'il faut faire AVANT d'activer IPS

Étape 1 : Observer (déjà réalisé quelques minutes avant)

- Mode IDS
- Analyse des alertes
- Identifier faux positifs

Étape 2 : Nettoyer les règles

Dans Snort :

- Désactive règles trop agressives
- Exemple :
 - Certaines règles web_server
 - Règles malware trop larges

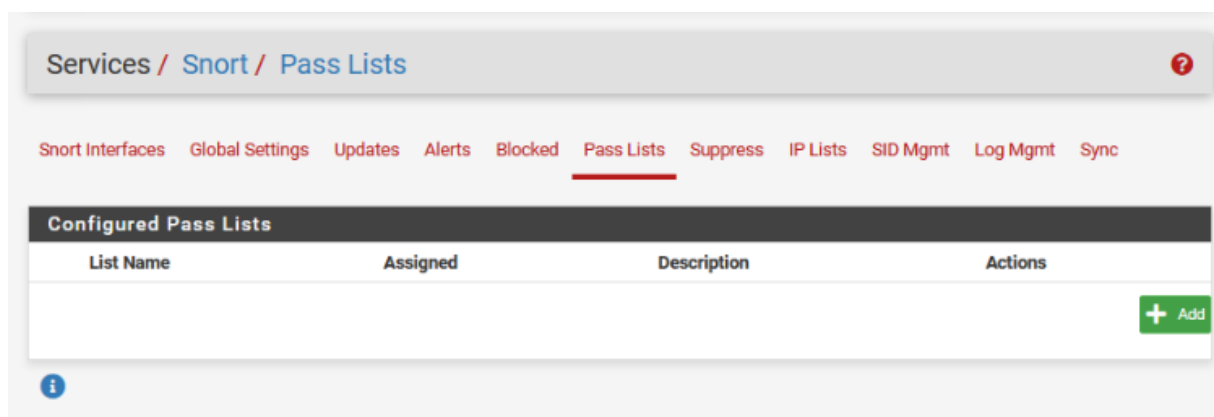
Il faut désactiver temporairement la règle suivante emerging server_web car elle est trop sensible en IPS.

☐ [emerging-web_server.rules](#)

Étape 3 : Whitelist

Ajoute :

- Ton IP admin



Snort InterfacesGlobal SettingsUpdatesAlertsBlockedPass ListsSuppressIP ListsSID MgmtLog MgmtSync

General Information

Namewhitelist_admin

The list name may only consist of the characters 'a-z, A-Z, 0-9 and _'.

Description

You may enter a description here for your reference.

Auto-Generated IP Addresses

Local Networks

☒ Add firewall Locally-Attached Networks to the list (excluding WAN). Default is Checked.

WAN Gateways

☒ Add WAN Gateways to the list. Default is Checked.

WAN DNS Servers

☒ Add WAN DNS servers to the list. Default is Checked.

Virtual IP Addresses

☒ Add Virtual IP Addresses to the list. Default is Checked.

VPN Addresses

☒ Add VPN Addresses to the list. Default is Checked.

Custom IP Addresses and Configured Firewall Aliases

Hint

Enter as many IP addresses or alias names as desired. Enter ONLY an IP address, IP subnet or alias name! Do NOT enter a FQDN (fully qualified domain name) directly! To use a FQDN, first create the necessary firewall alias, and then provide the alias name here. FQDN aliases are periodically re-resolved and updated by the firewall. You can also provide an IP subnet with a proper netmask of the form network/mask such as 1.2.3.0/24.

IP or Alias

192.168.30.0/24

Delete

Save

+ Add IP

3 nouvel

Snort InterfacesGlobal SettingsUpdatesAlertsBlockedPass ListsSuppressIP ListsSID MgmtLog MgmtSync

General Information

Namewhitelist_admin

The list name may only consist of the characters 'a-z, A-Z, 0-9 and _'.

Description

You may enter a description here for your reference.

Auto-Generated IP Addresses

Local Networks

☒ Add firewall Locally-Attached Networks to the list (excluding WAN). Default is Checked.

WAN Gateways

☒ Add WAN Gateways to the list. Default is Checked.

WAN DNS Servers

☒ Add WAN DNS servers to the list. Default is Checked.

Virtual IP Addresses

☒ Add Virtual IP Addresses to the list. Default is Checked.

VPN Addresses

☒ Add VPN Addresses to the list. Default is Checked.

Custom IP Addresses and Configured Firewall Aliases

Hint

Enter as many IP addresses or alias names as desired. Enter ONLY an IP address, IP subnet or alias name! Do NOT enter a FQDN (fully qualified domain name) directly! To use a FQDN, first create the necessary firewall alias, and then provide the alias name here. FQDN aliases are periodically re-resolved and updated by the firewall. You can also provide an IP subnet with a proper netmask of the form network/mask such as 1.2.3.0/24.

IP or Alias

LAN_ALL

Delete

Save

+ Add IP

1/4

Étape 4 : Choisir le bon mode

Dans pfSense :

- Legacy Mode → blocage via firewall

Etape 5 : Test depuis un client interne

```
Sélection Windows PowerShell
PS C:\Users\svc_admin> curl http://testmyids.com

StatusCode      : 200
StatusDescription : OK
Content         : uid=0(root) gid=0(root) groups=0(root)

RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  X-WS-Origin: available
                  X-WS-RateLimit-Limit: 1000
                  X-WS-RateLimit-Remaining: 999
                  Accept-Ranges: bytes
                  Content-Length: 39
                  Content-Type: text/html
                  Date: Thu...

Forms           : {}
Headers         : {[Connection, keep-alive], [X-WS-Origin, available], [X-WS-RateLimit-Limit, 1000],
                  [X-WS-RateLimit-Remaining, 999]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : System.__ComObject
RawContentLength : 39
```

Snort laisse passer le trafic normal

IPS fonctionne... MAIS

Il bloque seulement SI :

- Une règle match
- ET qu'elle est en mode DROP

```
PS C:\Users\svc_admin> nmap -sS -A scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-30 14:27 +0200
refresh_hostbatch: Failed to determine dst MAC address for target scanme.nmap.org (45.33.32.156)
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.75 seconds
PS C:\Users\svc_admin>
```

Maintenant :

Plus rien ne répond

Scan bloqué

IPS FONCTIONNE !

Attention, vérifier si nous même ne nous sommes pas bloqués.

```
PS C:\Users\svc_admin> ping 8.8.8.8

Envoi d'une requête 'Ping' 8.8.8.8 avec 32 octets de données :
Réponse de 8.8.8.8 : octets=32 temps=9 ms TTL=115
Réponse de 8.8.8.8 : octets=32 temps=10 ms TTL=115
Réponse de 8.8.8.8 : octets=32 temps=10 ms TTL=115

Statistiques Ping pour 8.8.8.8:
    Paquets : envoyés = 3, reçus = 3, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 9ms, Maximum = 10ms, Moyenne = 9ms
Ctrl+C
PS C:\Users\svc_admin> ping www.google.com

Envoi d'une requête 'ping' sur www.google.com [142.251.153.119] avec 32 octets de données :
Réponse de 142.251.153.119 : octets=32 temps=10 ms TTL=115
Réponse de 142.251.153.119 : octets=32 temps=11 ms TTL=115
Réponse de 142.251.153.119 : octets=32 temps=10 ms TTL=115
Réponse de 142.251.153.119 : octets=32 temps=10 ms TTL=115

Statistiques Ping pour 142.251.153.119:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 10ms, Maximum = 11ms, Moyenne = 10ms
```

Les tests réalisés ont démontré que le système SNORT configuré en mode IPS est capable de bloquer efficacement les scans réseau (Nmap) tout en autorisant le trafic légitime (ping, navigation web), ce qui valide le bon fonctionnement de la solution.

3. Schéma de ce qu'il a été mis en place

